Short Description of the Scene

This assignment demonstrates mipmapping and texture filtering techniques in real-time rendering using OpenGL. The objective was to analyze how different filtering modes affect visual quality, aliasing, and texture stability at varying distances and viewing angles.

The scene consists of a heavily tiled textured floor and a textured rotating sphere. A fully controllable keyboard-based camera allows observation of texture behavior at shallow angles and long distances, where mipmapping effects are most visible. An ImGui-based interface enables real-time switching between four filtering modes:

- No Mipmapping
- Nearest Mipmap
- Bilinear Mipmap
- Trilinear Filtering

Mipmaps are generated using glGenerateMipmap(), and filtering behavior is controlled via glTexParameteri() with different GL_TEXTURE_MIN_FILTER settings. The GPU automatically selects appropriate mip levels based on texture coordinate derivatives and performs the corresponding filtering. To further stress-test the filtering techniques, adjustable texture tiling is applied to the sphere, increasing texture frequency and making aliasing artifacts more apparent. This interactive implementation clearly demonstrates:

- Texture aliasing and undersampling
- Level-of-detail (LOD) transitions
- Mipmap level blending
- Quality vs performance trade-offs in texture sampling

Overall, the assignment highlights the importance of mipmapping in modern real-time rendering pipelines for improving visual stability, reducing shimmering artifacts, and optimizing texture bandwidth usage.